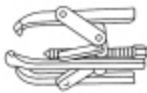


Crankshaft Pulley

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

 ST1184-A	3 Jaw Puller 303-D121 or equivalent
 ST2296-A	Installer, Front Cover Oil Seal 303-335
 ST1287-A	Replacer, Crankshaft Vibration Damper 303-102 (T74P-6316-B)
 ST1438-A	Strap Wrench 303-D055 (D85L-6000-A) or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Material

Item	Specification
Motorcraft® SAE 5W-30 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-30 Super Premium Motor Oil (Canada) XO-5W30-QSP (US); CXO-5W30-LSP12 (Canada)	WSS-M2C946-A

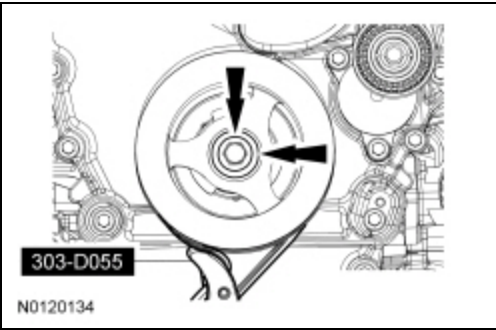
Removal

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to Section 100-02.
2. Remove the Air Cleaner (ACL) outlet pipe and the RH & LH ACL pipe-to-turbocharger pipes. Remove the LH turbocharger-to-Charge Air Cooler (CAC) tube and the CAC -to-Throttle Body (TB) pipe. For additional information,

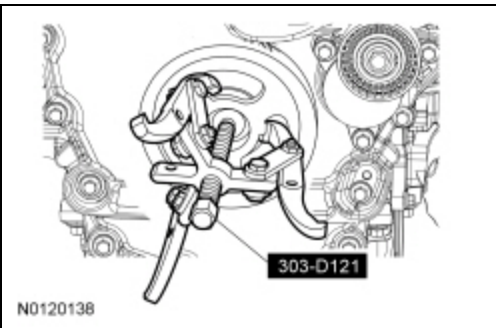
refer to Intake Air System Components — Exploded View, 3.5L Gasoline Turbocharged Direct Injection (GTDI) in Section 303-12.

3. Remove the accessory drive belt and the A/C belt. For additional information, refer to Section 303-05.

4. Using the Strap Wrench, remove the crankshaft pulley bolt and washer.
- Discard the bolt.



5. Using the 3 Jaw Puller, remove the crankshaft pulley.

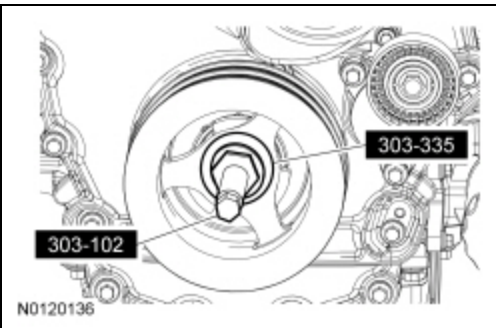


Installation

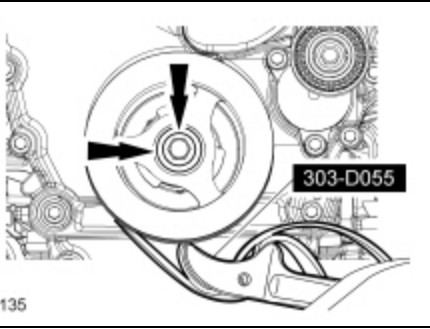
1. Lubricate the crankshaft front seal inner lip with clean engine oil.

2. **NOTE:** *Lubricate the outside diameter sealing surfaces with clean engine oil.*

Using the Front Cover Oil Seal Installer and Crankshaft Vibration Damper Replacer, install the crankshaft pulley.



3. Using the Strap Wrench, install the crankshaft pulley washer and new bolt and tighten in 4 stages.
- Stage 1: Tighten to 120 Nm (89 lb-ft).
 - Stage 2: Loosen one full turn.
 - Stage 3: Tighten to 50 Nm (37 lb-ft).
 - Stage 4: Tighten an additional 90 degrees.



4. Install the accessory drive belt and the A/C belt. For additional information, refer to Section 303-05.
5. Install the CAC -to-TB pipe and the LH turbocharger-to-CAC tube. Install the RH & LH ACL pipe-to-turbocharger pipes and the ACL outlet pipe. For additional information, refer to Intake Air System Components — Exploded View, 3.5L GTDI in Section 303-12.
6. After completing the repairs, use the scan tool to perform the Misfire Monitor Neutral Profile Correction procedure following the on-screen instructions.